

Date	07/07/26	Team ID	XXXXXX
Project Name	edugenie google gemini powered learning assistant		

Scalability & Future Plan

PROJECT EXPANSION & EVOLUTION ROADMAP

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team’s roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	Single-Instance Server Dependency The current backend runs on a single server instance with standard synchronous processing.	Can result in high latency, request timeouts, and server crashes during concurrent peak usage by multiple students.	High
2	API Rate Limits & Response Bottlenecks Direct synchronous calls to the external Google Gemini API are subject to rate limiting.	Failing to handle rate limits gracefully causes intermittent service interruptions and poor user experience during interactive chats.	High
3	Short-Term Chat Memory User context and session histories are stored locally or in temporary cache.	The assistant loses historical continuity and personalized progress context once a session expires or user logs out.	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Handles limited concurrent active connections before encountering response lag.	Deploy on Google Cloud Run or Kubernetes (GKE) with auto-scaling metrics to dynamically adjust to traffic spikes.
2	Data Storage	Simple relational or local file-based storage for quick metadata and prompt logs.	Migrate to Google Cloud Firestore or Bigtable for scalable, distributed NoSQL storage capable of handling massive telemetry.
3	Performance	Synchronous token generation waiting on deep API loops without active caching.	Implement Redis caching for frequently asked educational concepts and background message queues (Cloud Pub/Sub) for asynchronous tasks.
4	Security	Basic environment-level API key management and standard HTTPS connections.	Integrate Google Cloud IAM roles, robust OAuth2 user authentication, and strict PII-masking pipelines before sending data to LLM.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Multimodal Capabilities Integrate support for processing uploaded lecture videos, diagram images, and voice queries using Gemini's multimodal features.	Q4 2026	Allows students to learn via text, audio, and visual assets, drastically expanding accessibility.
Phase 3	Personalized Adaptive Learning Paths Develop analytics pipelines to tracks student learning performance and dynamically adapt future quiz generation.	Q2 2027	Provides hyper-customized study plans and detailed progress tracking dashboard for educators.
Phase 4	Edge AI Integration (Offline Mode) Incorporate on-device execution using lightweight models like Gemini Nano for basic offline text processing.	Q4 2027	Ensures uninterrupted learning experiences even in low-bandwidth or offline remote education zones.